

Serii numerice. Probleme propuse spre rezolvare

1. Studiați convergența și calculați suma pentru următoarele serii numerice:

$$\begin{aligned} \text{i)} & \sum_{n=1}^{\infty} \frac{1}{n(n+1)}; \text{ ii)} \sum_{n=1}^{\infty} \frac{1}{9n^2 + 3n - 2}; \text{ iii)} \sum_{n=3}^{\infty} \frac{1}{16n^2 - 8n - 3}; \\ \text{iv)} & \sum_{n=1}^{\infty} \frac{1}{\sqrt{3n+1} + \sqrt{3n-2}}; \text{ v)} \sum_{n=2}^{\infty} \frac{n^2 - 1}{(n+1)!}; \text{ vi)} \sum_{n=0}^{\infty} \arctan \frac{1}{n^2 + 3n + 3}; \\ \text{vii)} & \sum_{n=1}^{\infty} \left(\frac{5}{11}\right)^n; \text{ viii)} \sum_{n=2}^{\infty} \ln \left(1 - \frac{1}{n+1}\right); \text{ ix)} \sum_{n=1}^{\infty} \left(\frac{2001}{2020}\right)^n. \end{aligned}$$

2. Utilizând condiția necesară de convergență a unei serii să se arate că următoarele serii sunt divergente:

$$\begin{aligned} \text{i)} & \sum_{n=1}^{\infty} n \sin \frac{\pi}{4n}; \text{ ii)} \sum_{n=1}^{\infty} \left(\frac{2n}{2n+3}\right)^{n+1}; \text{ iii)} \sum_{n=1}^{\infty} \left(\frac{3n+1}{3n-1}\right)^{n^2}; \\ \text{iv)} & \sum_{n=1}^{\infty} n(2^{\sin \frac{1}{n}} - 1); \text{ v)} \sum_{n=1}^{\infty} \sin n^2; \text{ vi)} \sum_{n=1}^{\infty} (\sqrt{n^2 + n + 1} - \sqrt{n^2 - 1}); \\ \text{vii)} & \sum_{n=0}^{\infty} \arctan \frac{n^3 + 4n + 2}{n^2 + 3n + 1}; \text{ viii)} \sum_{n=1}^{\infty} \frac{\sin \frac{4}{n}}{\tan \frac{5}{n}}; \\ \text{ix)} & \sum_{n=1}^{\infty} \frac{1 \cdot 3 + 2 \cdot 4 + \dots + n(n+2)}{4n(1 + 3 + \dots + (2n-1))}; \text{ x)} \sum_{n=0}^{\infty} \frac{\sum_{k=1}^n (2k-1)^2}{\sum_{k=1}^n k(k+1)}. \end{aligned}$$

3. Utilizând criteriul fundamental al lui Cauchy, să se studieze convergența seriilor:

$$\text{i)} \sum_{n=1}^{\infty} \frac{\sin(n!x)}{n^2 + n}; \text{ ii)} \sum_{n=1}^{\infty} \frac{\cos(nx)}{n^2}; \text{ iii)} \sum_{n=1}^{\infty} \frac{\arctan(nx)}{2^n}, \quad x \in \mathbb{R}.$$

4. Utilizând criteriile de comparație să se studieze natura seriilor:

$$\text{i)} \sum_{n=1}^{\infty} \frac{1}{\sqrt{n^4 + n} - \sqrt{2n^4 - n}}; \text{ ii)} \sum_{n=1}^{\infty} n \sin \frac{\pi}{6n^4}; \text{ iii)} \sum_{n=1}^{\infty} \frac{\sqrt[3]{n}}{(n+1)\sqrt{n}};$$

$$\begin{aligned} \text{iv)} \sum_{n=1}^{\infty} \frac{1}{n+2020^n}; \text{ v)} \sum_{n=1}^{\infty} \frac{\sqrt{7n}}{n^2+3n+8}; \text{ vi)} \sum_{n=1}^{\infty} \frac{\ln(1+2020^n)}{2021^n}; \\ \text{vii)} \sum_{n=1}^{\infty} \frac{\sqrt[5]{3n^8+3n+2}}{9n^4+5n^3+4}; \text{ viii)} \sum_{n=1}^{\infty} \frac{n+\sqrt{n}}{2n^3-1} \text{ ix)} \sum_{n=1}^{\infty} \frac{4n^2-n+3}{n^3+2n}. \end{aligned}$$

5. Utilizând criteriul rădăcii, studiați natura seriilor:

$$\begin{aligned} \text{i)} \sum_{n=1}^{\infty} \left(\frac{n+1}{2n-1} \right)^n; \text{ ii)} \sum_{n=4}^{\infty} \left(\frac{2n+3}{2n-7} \right)^{n^2} \text{ iii)} \sum_{n=1}^{\infty} \frac{2n^{n+4}}{(4n+5)^n}; \\ \text{iv)} \sum_{n=1}^{\infty} \left(\alpha \cdot \frac{2n^2+n+1}{n^2} \right)^n; \alpha > 0; \text{ v)} \sum_{n=1}^{\infty} \left(\frac{n^2-5n+1}{n^2-4n+2} \right)^{n^2}; \\ \text{vi)} \sum_{n=1}^{\infty} (\sqrt{3n+2} - \sqrt{2n+1})^n; \text{ vii)} \sum_{n=1}^{\infty} \left(\frac{n^2+2n+3}{n^2+3n+2} \right)^{\frac{n^3}{2n+1}}; \\ \text{viii)} \sum_{n=1}^{\infty} (\sqrt{n(n+a)} - n)^n. \end{aligned}$$

6. Utilizând criteriul raportului, sau criteriul lui Raabe-Duhamel, studiați natura seriilor:

$$\begin{aligned} \text{i)} \sum_{n=2}^{\infty} \frac{8 \cdot 14 \dots (6n-4)}{9 \cdot 17 \dots (8n-7)}; \text{ ii)} \sum_{n=1}^{\infty} \frac{2^n}{n^2}; \text{ iii)} \sum_{n=1}^{\infty} \frac{n^n}{5^n n!}; \text{ iv)} \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!}; \\ \text{v)} \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n)!!} \frac{1}{2n+1}; \text{ vi)} \sum_{n=1}^{\infty} \frac{(2n-1)!}{(2n)!(2n+1)}; \text{ vii)} \sum_{n=1}^{\infty} n^2 \tan \frac{\pi}{2^n}; \\ \text{viii)} \sum_{n=1}^{\infty} (n^3+1) \sin \frac{\pi}{3^n}; \text{ vii)} \sum_{n=1}^{\infty} \frac{1}{\cosh n}; \text{ ix)} \sum_{n=1}^{\infty} \frac{(3n)^n}{(n+1)!}; \\ \text{x)} \sum_{n=1}^{\infty} \frac{n!}{(a+1)(a+2)\dots(a+n)}, \quad a > 0; \text{ xi)} \sum_{n=1}^{\infty} a^{\ln n}, a > 0. \end{aligned}$$